

Palm Oil Retention & Melting System

Submerged Hydronic Coil Design — Thermal/Throughput Analysis

Based on the *VCH Sizing Framework, Second Edition*
A Unified Design Methodology for Submerged Hydronic Coil Systems

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This document presents thermal sizing, heat loss analysis, NTU- ϵ performance evaluation, process control architecture, and production capacity validation for a 3×10 kL palm oil storage and 1×5 kL replenishment tank system, integrated with a passive solar thermal loop and auxiliary electric immersion heating.

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Executive Summary

This document presents a revised control architecture for the Nebico palm-oil storage facility, extending the original T_{10}/T_5 submerged hydronic coil design (VCH Sizing Framework, Second Edition) to accommodate a dedicated **palm olein** storage tank alongside the existing palm oil retention tanks. Each of the three 10 kL tanks ($T_{10,A}$ – $T_{10,C}$) now receives its own dedicated feed rather than a shared header, and one tank may be manually designated by site technicians as the **PO tank** (palm olein storage), which is then automatically and manually isolated from the common bunch and dispensed manually only. This restructuring required a new three-way priority architecture — T_5 trim-heating (P1), PO tank temperature maintenance (P2), and palm oil melting in the remaining T_{10} tanks (P3) — evaluated here under a constant 70 °C water supply, a 1.2 m s⁻¹ feed velocity cap, and a standardised 2 inch coil bore throughout (30 m per T_{10} /PO tank, 10 m for T_5).

Headline Results

Metric	Result
T_5 (P1) daily duty period	2.05 hr (8.5%)
PO tank (P2) daily duty period	0.59 hr (2.4%)
Palm oil melting (P3) daily window	21.37 hr (89.0%)
Palm oil melted per day, P3 window	≈2447 kg (2.45 t/day)
Fulfilment vs. 2 t/day mandate	≈ 122%
Total daily thermal energy demand	568.3 MJ
Average continuous power demand	6.58 kW
Solar (ETC) contribution to demand	≈40%
Auxiliary (36 kW) headroom vs. avg	> 5×

Key Findings

- **The revised architecture exceeds the original 2 t/day palm oil mandate by a comfortable margin (≈122%)**, even after diverting one full tank’s worth of storage capacity to palm olein and after accounting for T_5 and PO-tank heating demand ahead of it in priority.
- **The system is not heat-limited.** P1 and P2 together consume only 10.9% of the day’s coil availability; the true constraint on P3’s throughput is the ≈77.8 hr melt time of a full 10 kL tank, not available thermal power.
- **The PO tank’s own standing heat loss, not its sensible duty, dominates its energy budget** (roughly 7:1) because it remains uninsulated. Insulating the PO tank is identified as the single highest-leverage efficiency gain in this revision.
- **Priority arbitration between P1, P2, and P3 is handled by a sequential daily energy budget**, not a fixed-percentage flow split: P1 and P2 take what their duty periods require, and P3 receives the remainder of the 24-hour window, by design.

- **Fault detection is decoupled from duty-cycle bookkeeping.** Rather than alarming on elevated duty percentage, the system alarms on heating/cooling/loss rates that fall outside what is physically achievable given the coil and water conditions.

The remainder of this document develops the revised architecture, its governing logic, and the supporting thermal analysis from first principles.

1 Revised Storage and Control Architecture

1.1 From Shared Header to Individualised Feeds

The original T_{10} design drew all three retention tanks from a single shared water header, with tank selection handled entirely by the PLC's scoring logic (Section 11). This revision replaces the shared header with an **individualised feed per tank**: each of $T_{10,A}$, $T_{10,B}$, and $T_{10,C}$ receives its own branch, gated first by a manual ball valve and then by an automated control valve in series. This is the structural change that makes tank *individuality* possible — a prerequisite for designating one tank as a dedicated palm olein store without disturbing the piping or control logic of the other two.

1.2 The PO Tank

One T_{10} tank is designated by on-site technicians, via HMI input, to store palm olein rather than palm oil. Once designated, this tank — referred to throughout as the **PO tank** — is:

- **Automatically cut off** from the common T_{10} bunch: its automated valve is excluded from the P3 melting rotation and the legacy tank-scoring system (Section 11) no longer considers it a candidate for oil draw-off.
- **Manually cut off** as a physical safeguard: the ball valve ahead of the automated valve provides a hard, PLC-independent isolation point.
- **Dispensed manually only**, via the HMI-driven pulse logic described in Section 1.4, rather than through the priority-queued X/Y/Z distribution lines used for palm oil.

This isolation exists because palm olein and palm oil have materially different thermal behaviour: olein solidifies at $\approx 24^\circ\text{C}$ (versus palm oil's 36°C melting point) and, once liquid, requires comparatively little energy to keep that way. Left in the shared rotation, its low heat demand would distort the tank-scoring system's assumptions, which were derived entirely around palm oil's higher-energy melt behaviour.

1.3 The Three-Way Priority Ladder

The original two-tier logic (T_5 master, T_{10} demand-pull) is extended to three tiers:

Table 1. Revised priority architecture.

Tier	Process	Description
P1	T ₅ trim-heating	Unchanged in principle from the original design: highest priority, effectively guaranteed service.
P2	PO tank maintenance	New. Demand-variable, HMI-dispense triggered; bounded so it cannot starve P3 indefinitely.
P3	T ₁₀ palm oil melting	Unchanged heating logic, but now receives only the water-time remaining after P1 and P2 are satisfied.

This is deliberately not a hard-lockout priority scheme. A strict $P1 \gg P2 \gg P3$ lockout would mean P3 simply waits, indefinitely if necessary, whenever P1 or P2 has any demand at all. Because P3's task (palm oil melting) tolerates interruption far better than a live PO-tank dispense or T₅'s process-critical trim-heating, the architecture instead resolves to a **sequential daily energy budget**: P1's and P2's actual required duty periods are computed first, and P3 is allocated whatever remains of the 24-hour day. P3 is, by design, the process that absorbs all of the system's slack.

1.4 Why P2 Needed Its Own Sub-Architecture

Unlike P1 and P3, P2's demand is not a smooth, continuously-tracked thermal load — it is triggered by discrete, operator-initiated HMI dispense requests of variable size. The PO-tank dispense logic operates as follows:

1. The technician enters the quantity of oil to be dispensed via the HMI.
2. The PLC computes the time required for the dispense flow to deliver that quantity and runs the dispense valve for that duration.
3. Once the computed volume has been delivered, the system enters a **pulse-hold mode**: the coil's heating loop (not the oil dispense path) pulses on for a minimum of 20 s per 60 s window, keeping the residual oil in the branch warm while the technician completes the physical hand-off.
4. Pulse-hold continues until the technician registers "process complete" on the HMI, at which point the P2 heat demand is cleared entirely.

Because dispense events are operator-paced and unpredictable in timing, P2 cannot be scheduled the way P1's or P3's loads can. This is why P2 is evaluated here on a fixed **daily throughput assumption** (500 kg/day) rather than a real-time trigger pattern — the resulting duty period represents the total daily water-time P2 requires, not the timing of individual dispense events.

2 Fault Detection: A Physics-Based Alarm Philosophy

An earlier iteration of this architecture considered raising an alarm whenever P1’s duty cycle, measured over a short rolling window, exceeded twice its statistically expected value. This was rejected for a specific reason: **P1’s real-world demand is inherently bursty** — it activates only when θ_S drops below its lower setpoint, so legitimate operation will naturally produce some windows with heavy, clustered P1 activity and others with none. A duty-percentage threshold cannot distinguish a legitimately busy stretch of the day from an actual fault (a stuck valve, a drifting thermometer, a failed sensor), so a window-based trigger risks either false alarms on busy days or missed faults on quiet ones.

Revised philosophy: alarm on physics, not on bookkeeping. Instead of watching duty-cycle percentage, the PLC compares T_5 ’s *observed* heating and cooling rates against the *maximum physically achievable* rates given the coil’s known heat-delivery capacity and the tank’s own computed loss rate:

- If T_5 is observed heating *faster* than $\dot{Q}_{T_5}/(m C_p)$ could physically produce at the current water supply temperature, the reading is inconsistent with reality — sensor or valve fault.
- If T_5 is observed cooling *faster* than its own standing loss rate ($UA \Delta T$, Section 5) can account for, that too is physically inconsistent — a leak, a stuck-open coil valve, or a sensor fault.

Under normal high-demand conditions, P1 is still permitted to run over its statistically expected duty period without triggering the alarm, because doing so is not a fault; it is simply a busier-than-average day.

A secondary, purely structural safeguard remains in place: the arbitration window used to size P2’s allocation (Section 7) is itself sized off T_5 ’s own worst-case heat-up time, so that no single window is so short it misrepresents a legitimate heating event as anomalous, nor so long that a genuine fault goes undetected for an unreasonable period.

3 Revised System Parameters

Table 2. Parameters changed from the original T_{10}/T_5 design.

Parameter	Original	Revised
Feed pipe	1-inch Sch. 40 CPVC	Unchanged (1-inch)
Design feed velocity	1.5 m s^{-1}	1.2 m s^{-1}
Coil bore	1–2 inch (variable, solved)	2 inch (fixed, all tanks)
T_{10} coil length	Solved ($\approx 56\text{--}90 \text{ m}$)	Fixed at 30 m
T_5 coil length	Solved ($\approx 3\text{--}6 \text{ m}$)	Fixed at 10 m
Water supply temperature	$70\text{--}75^\circ\text{C}$ (variable)	Constant 70°C
Tank feed topology	Shared header	Individual feed per tank

4 Water-Side Hydraulics (Shared Basis)

4.1 Flow Rate and Water Capacity Rate

Water properties at $T_{\text{hot}} = 70^\circ\text{C}$: $\rho_w = 977.8 \text{ kg m}^{-3}$, $C_{p,w} = 4190 \text{ J kg}^{-1} \text{ K}^{-1}$, $\mu_w = 4.04 \times 10^{-4} \text{ Pa s}$, $k_w = 0.663 \text{ W m}^{-1} \text{ K}^{-1}$, $\text{Pr}_w = 2.55$.

At the revised design velocity $V_{\text{max}} = 1.2 \text{ m s}^{-1}$, 1-inch supply ($d_{i,\text{sup}} = 26.64 \text{ mm}$):

$$\dot{m}_w = \rho_w \frac{\pi d_{i,\text{sup}}^2}{4} V_{\text{max}} = 977.8 \times \frac{\pi \times 0.02664^2}{4} \times 1.2 = 0.654 \text{ kg s}^{-1} \quad (1)$$

$$C_w = \dot{m}_w C_{p,w} = 0.654 \times 4190 = 2740.7 \text{ W K}^{-1} \quad (2)$$

Since P1, P2, and P3 operate one branch at a time (this document evaluates the sequential-branch case only; simultaneous-branch operation is discussed qualitatively in Section 10), the full capacity rate C_w is available to whichever coil is active.

4.2 Batch-Side Modelling: Why $C_{\text{min}} = C_w$ Here

Modelling note. The original T₅ analysis treated the *oil throughput rate* as C_{min} , appropriate for a continuous flow-through duty. Under the revised architecture, each tank is heated as a well-mixed *batch* whose bulk temperature changes slowly relative to a single pass of water through the coil. The correct model is therefore the opposite: the **water side is C_{min}** ($C_w = 2740.7 \text{ W K}^{-1}$), and the tank contents behave as $C_{\text{max}} \rightarrow \infty$ over the timescale of one coil pass. Using the throughput-rate model here would predict effectiveness in excess of 98% and near-instant overshoot past the target temperature band — physically inconsistent with a duty-cycled batch process. All NTU calculations in this document therefore use:

$$NTU = \frac{UA}{C_w}, \quad \varepsilon = 1 - e^{-NTU}, \quad Q(T_{\text{tank}}) = \varepsilon C_w (T_{\text{hot}} - T_{\text{tank}}) \quad (3)$$

4.3 Coil-Side Flow — 2-inch Bore (All Tanks)

With the coil bore now fixed at 2-inch NPS Sch. 10S ($d_i = 54.79 \text{ mm}$, $d_o = 60.33 \text{ mm}$) for every tank:

$$Re_{\text{coil}} = \frac{4 \dot{m}_w}{\pi d_i \mu_w} = \frac{4 \times 0.654}{\pi \times 0.05479 \times 4.04 \times 10^{-4}} \approx 37,626 \quad (\text{turbulent}) \quad (4)$$

Gnielinski correlation with Kubair–Kuloor helical enhancement:

$$f = (0.79 \ln Re_{\text{coil}} - 1.64)^{-2} = 0.0224 \quad (5)$$

$$Nu_{\text{str}} = \frac{(f/8)(Re_{\text{coil}} - 1000) \text{Pr}_w}{1 + 12.7 \sqrt{f/8} (\text{Pr}_w^{2/3} - 1)} = 165.2 \quad (6)$$

Table 3. Coil-side inner heat transfer coefficient by tank, 2-inch bore.

Parameter	T_{10} / PO tank ($D_{\text{helix}} = 2.072$ m)	T_5 ($D_{\text{helix}} = 1.711$ m)
$Nu_{\text{helix}} = Nu_{\text{str}}(1 + 3.54 d_i/D_{\text{helix}})$	180.7	183.9
$h_i = Nu_{\text{helix}} k_w/d_i$ (W/m ² .K)	2187	2227

4.4 Overall U -Values

Using the same fouling allowances and wall-conduction basis as the original VCH framework, evaluated at the 2-inch bore ($R_{\text{wall}} = 1.782 \times 10^{-4}$, $R_{f,i} \cdot d_o/d_i = 1.10 \times 10^{-4}$, $R_{f,o} = 9.0 \times 10^{-4}$):

Table 4. Overall U -values used in the revised architecture.

Duty	Basis	U (W/m ² .K)
T_5 trim-heat (liquid, $h_o = 138.9$)	10 m coil	112.6
PO tank maintenance (liquid, $h_o = 138.9$)	30 m coil	112.5
T_{10} palm oil melt (solid, $h_o = 35$)	30 m coil	33.04

As in the original study, the water-side resistance is a small fraction of the total for the melting duty — the outer oil-side film continues to dominate $1/U$ once the phase is solid. This is why fixing the coil bore at 2-inch (rather than continuing to treat bore as a free variable) costs almost nothing in thermal performance while simplifying procurement to a single pipe spec across every tank.

5 P1 — T₅ Duty Cycle Analysis

5.1 Coil Delivery Rate

T₅ coil: $A = \pi d_o L = \pi \times 0.06033 \times 10 = 1.895 \text{ m}^2$.

$$NTU_{T5} = \frac{U_{T5} A}{C_w} = \frac{112.6 \times 1.895}{2740.7} = 0.0779 \Rightarrow \varepsilon_{T5} = 1 - e^{-0.0779} = 0.0749 \quad (7)$$

$$\varepsilon_{T5} C_w = 205.3 \text{ W K}^{-1} \Rightarrow Q_{T5}(T) = 205.3 \times (70 - T) \text{ W} \quad (8)$$

$$Q_{T5}(40^\circ\text{C}) = 6160 \text{ W} \quad Q_{T5}(45^\circ\text{C}) = 5133 \text{ W} \quad (9)$$

5.2 Daily Energy Demand

T₅ operates on a fixed 2 t/day throughput, heated from an intake of 40 °C to a process ceiling of 45 °C, plus its own standing jacket loss (insulated per the original design, $U \approx 0.686 \text{ W/m}^2\cdot\text{K}$, $A = 14.25 \text{ m}^2$, evaluated at the mid-band $\Delta T = 24.4 \text{ K}$).

$$E_{\text{sensible}} = 2000 \times 2100 \times 5 = 21.0 \text{ MJ} \quad (10)$$

$$E_{\text{loss}} = U A \Delta T \times 86,400 \text{ s} = 0.686 \times 14.25 \times 24.4 \times 86,400 = 20.62 \text{ MJ} \quad (11)$$

$$E_{P1} = 21.0 + 20.62 = 41.62 \text{ MJ/day} \quad (12)$$

5.3 Duty Period

At the mid-band condition (42.5 °C), $Q_{T5} = 205.3 \times 27.5 = 5645.8 \text{ W}$.

$$t_{P1} = \frac{41.62 \times 10^6}{5645.8} = 7371 \text{ s} \approx \mathbf{2.05 \text{ hr (8.5\% of 24 hr)}} \quad (13)$$

6 P2 — PO Tank Duty Cycle Analysis

6.1 Coil Delivery Rate

PO tank shares T₁₀ geometry: $A = \pi \times 0.06033 \times 30 = 5.685 \text{ m}^2$.

$$NTU_{PO} = \frac{112.5 \times 5.685}{2740.7} = 0.2334 \Rightarrow \varepsilon_{PO} = 0.2081 \quad (14)$$

$$\varepsilon_{PO} C_w = 570.5 \text{ W K}^{-1} \Rightarrow Q_{PO}(T) = 570.5 \times (70 - T) \text{ W} \quad (15)$$

$$Q_{PO}(24^\circ\text{C}) = 26.24\text{ kW} \quad Q_{PO}(30^\circ\text{C}) = 22.82\text{ kW} \quad (16)$$

6.2 Daily Energy Demand

Assumed daily olein throughput: 500 kg/day, held in the 24–30 °C band (literature $C_{p,\text{olein}} \approx 2100\text{ J kg}^{-1}\text{ K}^{-1}$, consistent with the liquid palm oil value used elsewhere in the VCH framework). The PO tank remains **uninsulated** (mild steel, same geometry as T_{10}), evaluated at its mid-band condition (27 °C, $\Delta T = 8.9\text{ K}$):

$$E_{\text{sensible}} = 500 \times 2100 \times 6 = 6.30\text{ MJ} \quad (17)$$

$$\begin{aligned} E_{\text{loss}} &= (U_{\text{cyl}}A_{\text{cyl}} + U_{\text{base}}A_{\text{base}})\Delta T \times 86,400\text{ s} \\ &= (3.32 \times 16.035 + 1.38 \times 4.229) \times 8.9 \times 86,400 = 45.4\text{ MJ} \end{aligned} \quad (18)$$

$$E_{P2} = 6.30 + 45.4 = 51.7\text{ MJ/day} \quad (19)$$

The PO tank's loss term outweighs its sensible duty by roughly 7:1. Because it sits ahead of P3 in priority, this uninsulated loss is effectively taxing the melting window before palm oil ever sees water. Insulating the PO tank to the same standard as T_5 would cut E_{P2} substantially and directly return that time to P3 — the single highest-leverage change identified in this analysis.

6.3 Duty Period

At the mid-band condition, $Q_{PO} = 570.5 \times 43 = 24\,531.5\text{ W}$.

$$t_{P2} = \frac{51.7 \times 10^6}{24,531.5} = 2107\text{ s} \approx \mathbf{0.585\text{ hr (2.4\% of 24 hr)}} \quad (20)$$

7 Arbitration Window Sizing

The 10-minute arbitration window used to bound P2's allocation (Section 1.4) is sized as three times the time required to heat a full T_5 charge from 40 °C to 45 °C under worst-case loss — long enough that a single legitimate T_5 heating event does not itself trip the window, short enough that P2 cannot be shut out for an unreasonable stretch of the day. Within each window, P2 is capped at a flat **5 of every 10 minutes**, independent of whether P1 or P3 happen to be active; P1 always preempts on demand (subject to the physics-based fault check of Section 2), and P3 receives whatever remains.

8 P3 — Palm Oil Melting Window

8.1 Residual Window

$$t_{P3} = 24 \text{ hr} - (t_{P1} + t_{P2}) = 24 - 2.633 \approx \mathbf{21.37 \text{ hr (89.0\% of 24 hr)}} \quad (21)$$

8.2 Coil Delivery Rate (Solid-Phase, 2-inch Bore, 30 m)

$$NTU_{T10} = \frac{33.04 \times 5.685}{2740.7} = 0.0685 \Rightarrow \varepsilon_{T10} = 0.0662 \quad (22)$$

$$Q_{T10}(36^\circ\text{C}) = 0.0662 \times 2740.7 \times (70 - 36) \approx 6172 \text{ W} \quad (23)$$

8.3 Melt Energy to a 40 °C Delivery Target

Extending the base VCH specific-energy formula with a small liquid-phase sensible tail above the melting point, so that oil is delivered to T₅'s 40 °C intake rather than left exactly at its 36 °C melting point:

$$q_{40} = C_{p,s} (36 - 18.1) + L_f + C_{p,l} (40 - 36) = 35,800 + 150,000 + 8,400 = 194.2 \text{ kJ kg}^{-1} \quad (24)$$

For a full 10 kL tank ($m \approx 8900 \text{ kg}$, per the original energy budget):

$$E_{\text{full tank}} = 8900 \times 194,200 = 1728.4 \text{ MJ} \Rightarrow t_{\text{full melt}} = \frac{1728.4 \times 10^6}{6172} \approx 77.8 \text{ hr} \quad (25)$$

A single tank, run continuously at this rate, takes 77.8 hr to fully melt — confirming, as in the original study, that melt time (not available heat) is the system's real bottleneck, and that throughput depends on staggering multiple tanks through their melt cycles rather than waiting on any one tank to finish.

8.4 Mass Melted Within the P3 Window

$$E_{P3} = 6172 \text{ W} \times (21.37 \times 3600) \text{ s} \approx 475.0 \text{ MJ} \quad (26)$$

$$m_{P3} = \frac{475.0 \times 10^6}{194,200} \approx \mathbf{2447 \text{ kg/day (2.45 t/day)}} \quad (27)$$

9 Aggregate Daily Demand and Supply Fulfilment

9.1 Total Daily Thermal Energy Demand

$$E_{\text{total}} = E_{P1} + E_{P2} + E_{P3} = 41.62 + 51.7 + 475.0 = 568.3 \text{ MJ/day} \quad (28)$$

$$P_{\text{avg}} = \frac{568.3 \times 10^6}{86,400} \approx 6.58 \text{ kW continuous average} \quad (29)$$

9.2 Supply-Side Fulfilment

Against the existing supply — the $3 \times 12 \text{ kW}$ (36 kW) auxiliary immersion battery and the ETC solar configuration (228.4 MJ/day from the base VCH proposal):

Table 5. Aggregate demand vs. available supply.

Quantity	Value
Total daily thermal demand	568.3 MJ
ETC solar daily yield	228.4 MJ
Solar coverage of daily demand	$\approx 40\%$
Auxiliary capacity (24 hr $\times$ 36 kW)	864 MJ
Auxiliary headroom vs. average 6.58 kW demand	$> 5\times$

Supply fulfilment: 100% of thermal demand, with solar (ETC) covering $\approx 40\%$ of the daily energy requirement and auxiliary capacity comfortably backstopping the remainder — the system is not heat-supply constrained under this architecture.

9.3 Throughput Fulfilment vs. the 2 t/day Mandate

Taking the original 2 t/day palm oil target as the 100% baseline, and P3's actual melted output as the delivered figure:

$$\text{Fulfilment} = \frac{2447 \text{ kg}}{2000 \text{ kg}} \times 100\% \approx \mathbf{122\%} \quad (30)$$

The revised architecture exceeds the original 2 t/day palm oil mandate by roughly 22%, despite dedicating one full T_{10} tank to palm olein storage and despite P1 and P2 taking priority ahead of it — a direct consequence of the melt-time bottleneck being far less sensitive to the small (10.9% combined) water-time taken by P1 and P2 than it is to raw available melt duration.

10 Architectural Outlook: Toward Simultaneous Branch Operation

All analysis in this document assumes **sequential, single-branch** operation: only one of P1, P2, or P3 draws water at any instant, and the full capacity rate C_w is available to whichever is active. This was the conservative, easier-to-verify basis for establishing the baseline throughput figures above. However, the coil sizing itself — 2-inch bore throughout, with Q_{target} margins well above what P1 and P2 actually require — suggests the trunk main is not the limiting element, and a further architectural evolution is possible.

10.1 The Re-Evaluation Concept (Not Calculated Here)

Rather than routing the full trunk flow down a single branch at a time, the trunk main itself could be upsized to a larger bore and higher flow rate, while the individual tank branches remain as specified. This would allow **multiple branches to be served simultaneously** rather than strictly sequentially. The priority order is preserved, but its mechanism changes:

- **P3 becomes a bypass branch**, permanently valved into the loop rather than gated behind P1/P2 completion.
- **Whenever P1 (T_5) is in its duty cycle**, its valve and the currently-selected T_{10} tank's valve can both remain open simultaneously, letting the upsized trunk flow split between them rather than forcing P3 to wait out the entirety of every P1 event.
- P2 would need its own concurrency rule against this shared trunk, analogous to the flow-sharing logic considered (and ultimately simplified away) during the development of this architecture.

This would shorten P3's effective waiting time per P1 event and push the system's throughput closer to its coil-delivery ceiling rather than its arbitration-window ceiling. It is presented here as a design direction for a future revision; no flow-split, trunk-sizing, or duty-cycle math has been carried out for this concept, consistent with the scope of the present analysis.

11 Legacy Control Elements Retained from the Original Design

The following elements of the original T_{10}/T_5 control architecture are retained unchanged, and continue to govern operation among the two remaining (non-PO) T_{10} tanks:

- The tank selection scoring system (S_{casual} , $S_{\text{immediate}}$), now applied only across the two tanks not designated as the PO tank.
- The dynamic upper thermal boundary function $T_f(x, y, m_p)$ governing T_5 's safety cut-off logic.
- The three solar circulation modes (through-solar, bypass-solar, quick-solar cycle) and the thermal battery recharge interlock.

12 System Estimation and Conclusions

1. **The revised architecture is a net improvement on the original 2 t/day mandate**, delivering an estimated **2.45 t/day** of melted palm oil ($\approx 122\%$ fulfilment) even after committing one full tank to palm olein storage and after P1/P2 take their priority share of water-time.
2. **The system runs well under its thermal ceiling**. Combined P1+P2 duty is only 10.9% of the day; P3's real constraint is melt time (77.8 hr per full tank), not coil heat delivery, and aggregate demand (6.58 kW average) sits far below available supply (36 kW auxiliary alone provides $> 5\times$ headroom).
3. **The PO tank should be insulated**. Its loss term (≈ 45.4 MJ/day) outweighs its sensible duty (≈ 6.3 MJ/day) by roughly 7:1; bringing it to T_5 's insulation standard would shrink t_{P2} and hand that time back to P3 at effectively no capital cost beyond the insulation itself.
4. **The sequential daily-budget model for P1/P2/P3 is simple, auditable, and sufficient at current demand levels**. The 10-minute arbitration window and flat 5-minute P2 cap provide a bounded, predictable ceiling on P2 without requiring per-window renegotiation of priority.
5. **Fault detection should remain physics-based, not duty-cycle based**. Comparing observed T_5 heating/cooling rates against the coil's known delivery capacity and the tank's own loss rate distinguishes genuine faults from ordinary busy-day variation, avoiding both false alarms and missed detections.
6. **The single largest available headroom for future throughput growth is the melt-time bottleneck, not heat supply**. Given supply headroom exceeds $5\times$ current average demand, the simultaneous-branch concept outlined in Section 10 — rather than any further increase in auxiliary or solar capacity — is the recommended direction for a subsequent revision.

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